

Amar Mahato

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PROFILE

Software Developer with experience building systems across event-driven pipelines, multi-agent architectures, and developer tooling. Delivered systems in **TypeScript, Python, and Go** — including a LangGraph multi-agent targeting system with LangSmith observability, an LLM-powered meeting assistant with RAG retrieval, and an MCP server for LLM-driven binary analysis. Experienced in distributed systems (RabbitMQ), vector databases (Qdrant), and cloud deployment (Docker, AWS).

PROJECTS

- Concize** | *Node.js, RabbitMQ, Cerebras, Sarvam AI, Groq, Gemini, Qdrant, MongoDB* [GitHub](#)
- Engineered the backend for a **Chrome Extension AI meeting assistant** (all web meeting platforms) that lets users chat with meeting content via a RAG chatbot; captures periodic audio chunks rather than continuous streaming to reduce STT costs.
 - Designed a **dual-storage RAG pipeline**: raw transcriptions stored verbatim in MongoDB (source of truth) while a clean service resolves chunk-local speaker ambiguity (diarization IDs reset per chunk) by converting dialogue to anonymized 3rd-person narratives before embedding into Qdrant.
 - Solved global timeline reconstruction across async-uploaded chunks via a custom **x-audio-offset** header: each chunk's relative timestamps are shifted by the client-tracked global offset before storage, ensuring accurate meeting-wide positioning across both databases.
 - Hardened the pipeline for production with atomic MongoDB locking for sequential consistency, multi-key LLM API rotation with exponential backoff, prompt guardrails, and a 95-test suite with 100% pass rate.
- Trecl** | *Python, LangGraph, LangSmith, Qdrant, Gemini, Cerebras, Tavily, PyGitHub, Pydantic* [GitHub](#)
- Built an **autonomous company targeting system** in LangGraph: input is a company name, user stack, and anti-persona constraints; 6 agents research the org across web, GitHub, and job boards — classifying entry vectors into 3 tiers (active listings, OSS contribution pitches, cold outreach with custom project proposal) and generating tailored outreach per target — suspending via `interrupt_before` for human selection (SQLite-checkpointed).
 - Designed the **GitHub analyst as a compiled ReAct sub-graph** with isolated state (preventing tool-call message bloat leaking into parent state), 7 tools, and a **4-layer stateful guardrail system**: module-level registries enforce tool-call ordering, reject hallucinated repo names, cache labels before filtering, sanitize queries against banned GitHub Search operators, and an AI message counter caps the loop at 15 to avoid LangGraph's recursion limit.
 - Implemented **task-typed agentic RAG**: Tavily `site:-`targeted scraping indexed into Qdrant with Gemini embeddings (RETRIEVAL.DOCUMENT/RETRIEVAL.QUERY task types) and payload indexes on `source_type/timestamp_epoch` for server-side filtered retrieval; hardened with tenacity retry proxy (429/5xx-only), round-robin key rotation across 4 services, and a 4-tier pytest suite with MockLLM for API-free integration tests.
- Headless Ghidra MCP** | *Python, Java, Ghidra, MCP, Capstone, pefile* [GitHub](#)
- Architected a **Cross-Platform MCP Server** enabling LLMs to perform automated binary analysis, decompilation, call-graph traversal, and malware triage without Ghidra's GUI.
 - Engineered a **Python-Java Bridge** with a custom Ghidra script serializing functions, strings, imports, and cross-references into cached JSON; hardened with LRU session caching, hybrid streaming for large binaries, path-traversal security, and a 60-test suite (Windows/Linux).

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, SQL, HTML/CSS
Backend & APIs: Node.js, Express.js, FastAPI, Socket.io, RabbitMQ, REST APIs, JWT, OAuth2
Applied AI: RAG Pipelines, AI Agents, LangGraph, LangSmith, MCP, Qdrant, Whisper V3, Prompt Engineering, Gemini/Groq/Cerebras SDKs
Database & Storage: PostgreSQL, MongoDB, Redis, Supabase, Prisma ORM, Convex, Qdrant
Frontend: React.js, Next.js, Tailwind CSS, WebSockets, Chrome Extension API
Tools and Deployment: Docker, GitHub Actions (CI/CD), AWS (EC2, S3), Linux, Jest, Render, Vercel

EDUCATION

Dr. B. C. Roy Engineering College <i>Bachelor of Technology, Computer Science and Engineering</i>	Durgapur, West Bengal, India
Beachwood School <i>Higher Secondary</i>	Durgapur, West Bengal, India